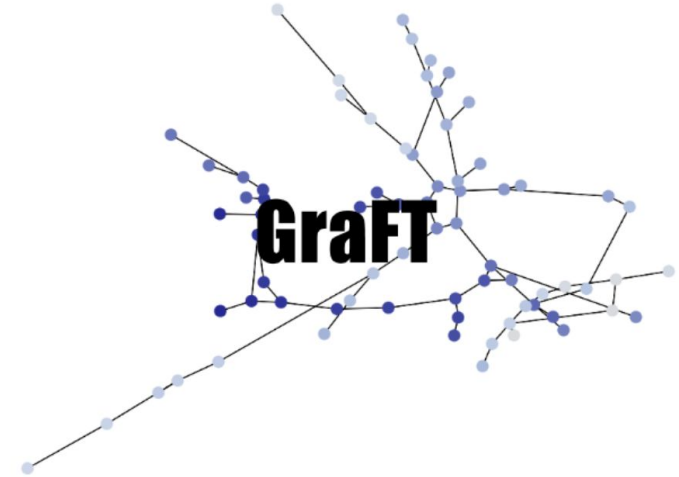


Refining Network-Based Framework for Actin Filament Identification

Presentation by Senyu Zhu
Supervisor: Dr. Jacqueline Nowak

GraFT: Graph of Filaments over Time

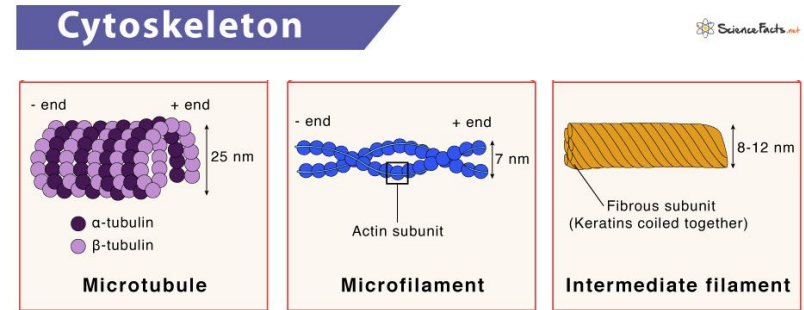
- GraFT was developed by Dr. Isabella Østerlund
- This framework combines the concepts of image analysis and graph theory
- Traces and tracks filamentous structures such as the actin filaments



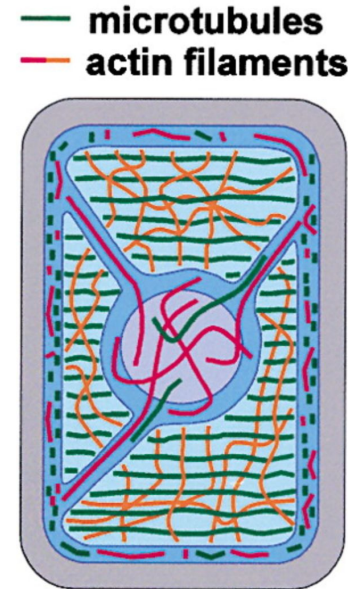
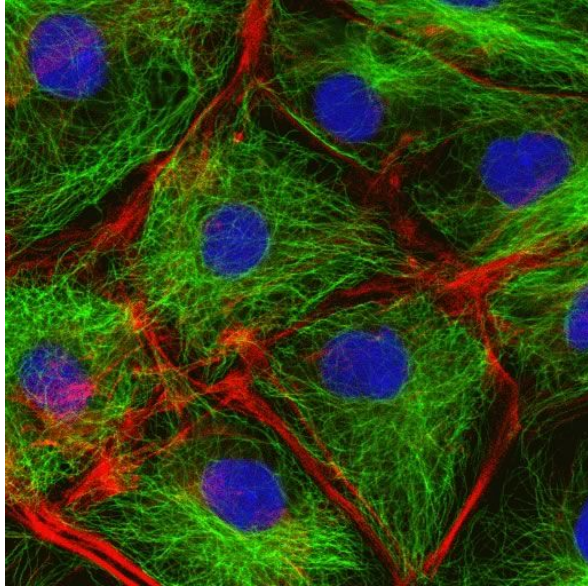
(Nikoloski et. al, 2024)

Motivation: Why are we developing this framework?

- Function of the cytoskeleton:
 - Maintains the shape and internal organization
 - Provides mechanical support
 - Enables cellular movement
 - Facilitates intracellular transport
- Study their properties and organization using the GraFT framework



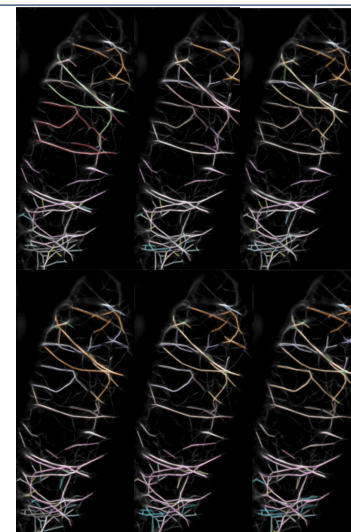
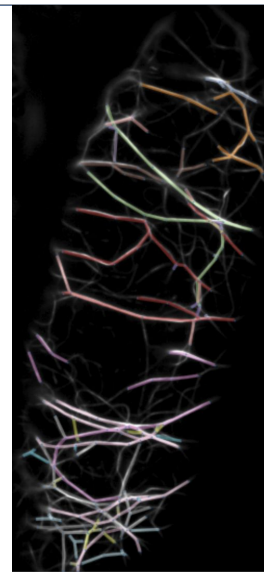
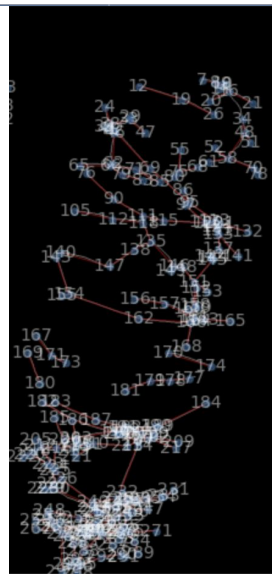
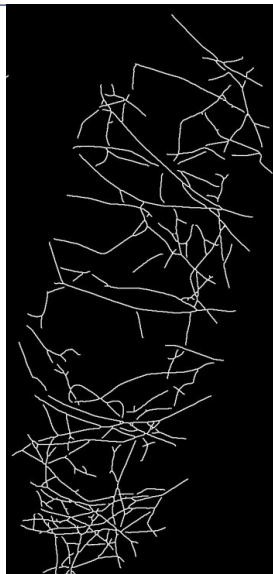
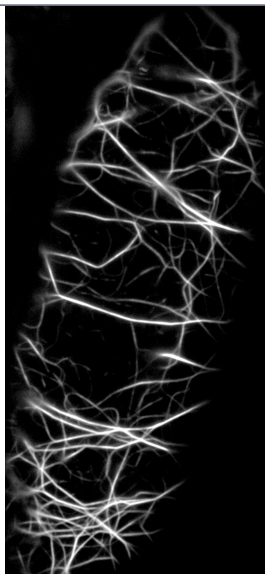
Cytoskeleton in plant cells – limitations in actin filaments



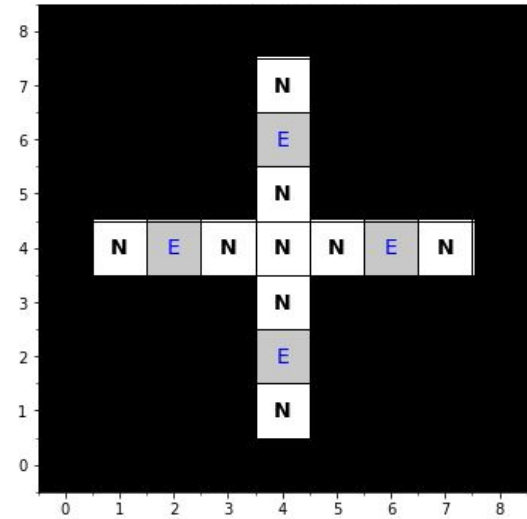
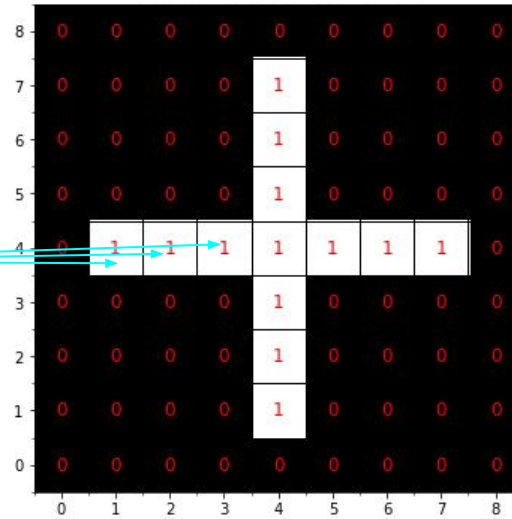
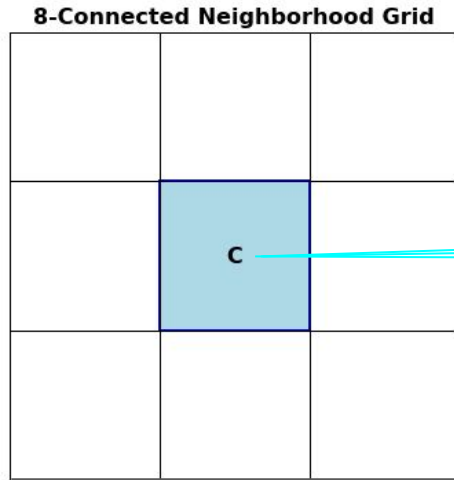
Actin filaments are more flexible and dynamic than microtubules

(Kost & Chua, 2002)

GraFT: Workflow



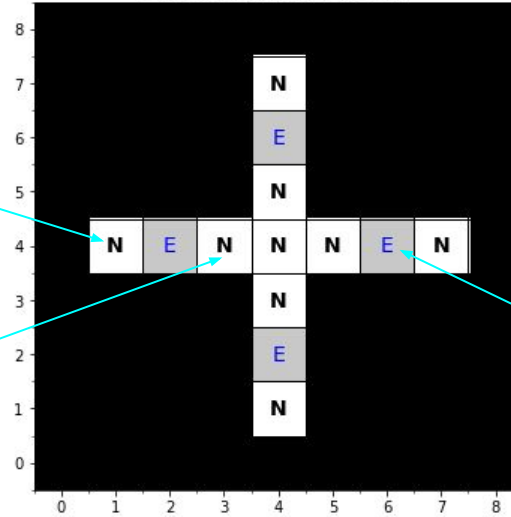
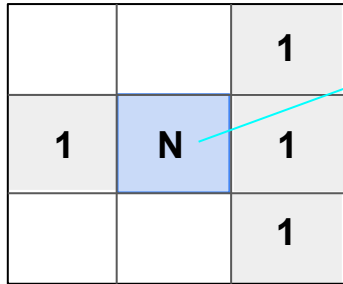
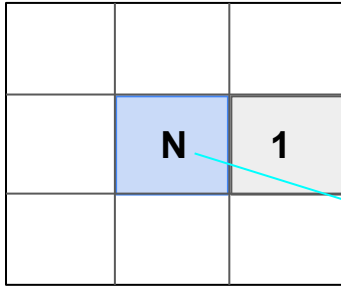
GraFT: Node placement



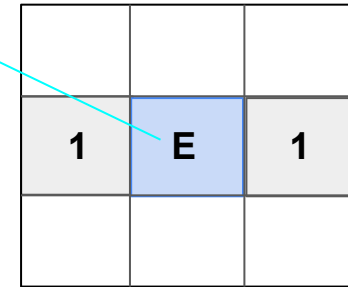
Kernel iterates through structure

GraFT: Node placement

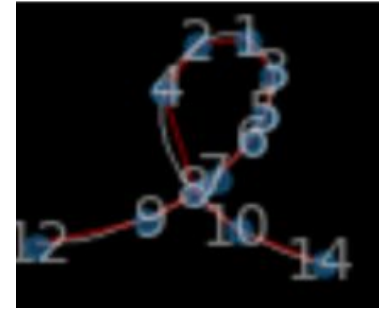
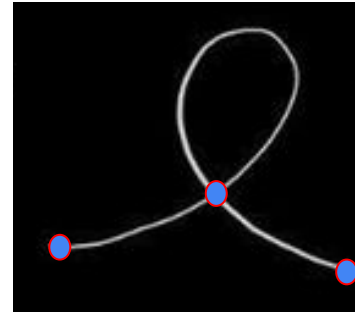
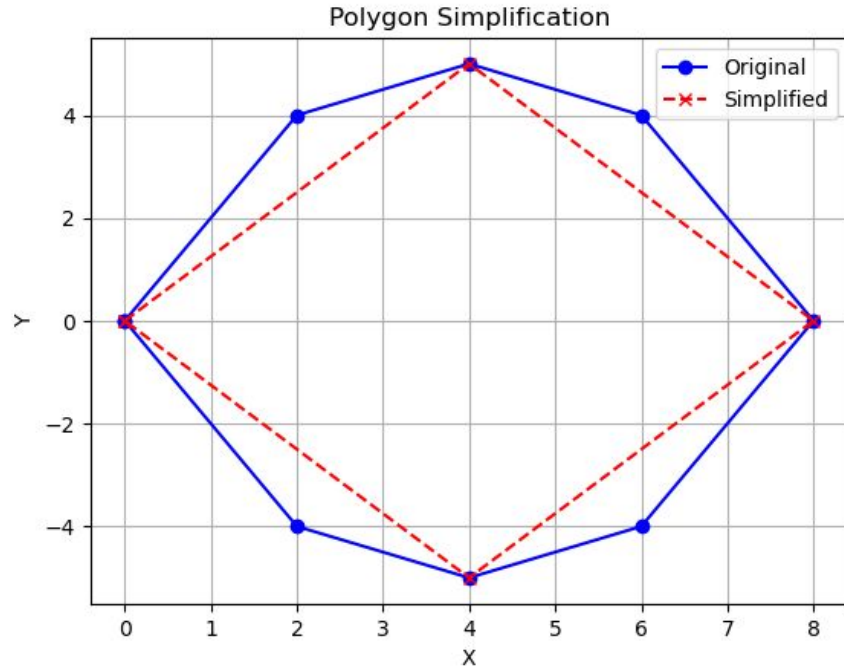
Center pixel has 1 or 3 or more neighbors → Dangling Node



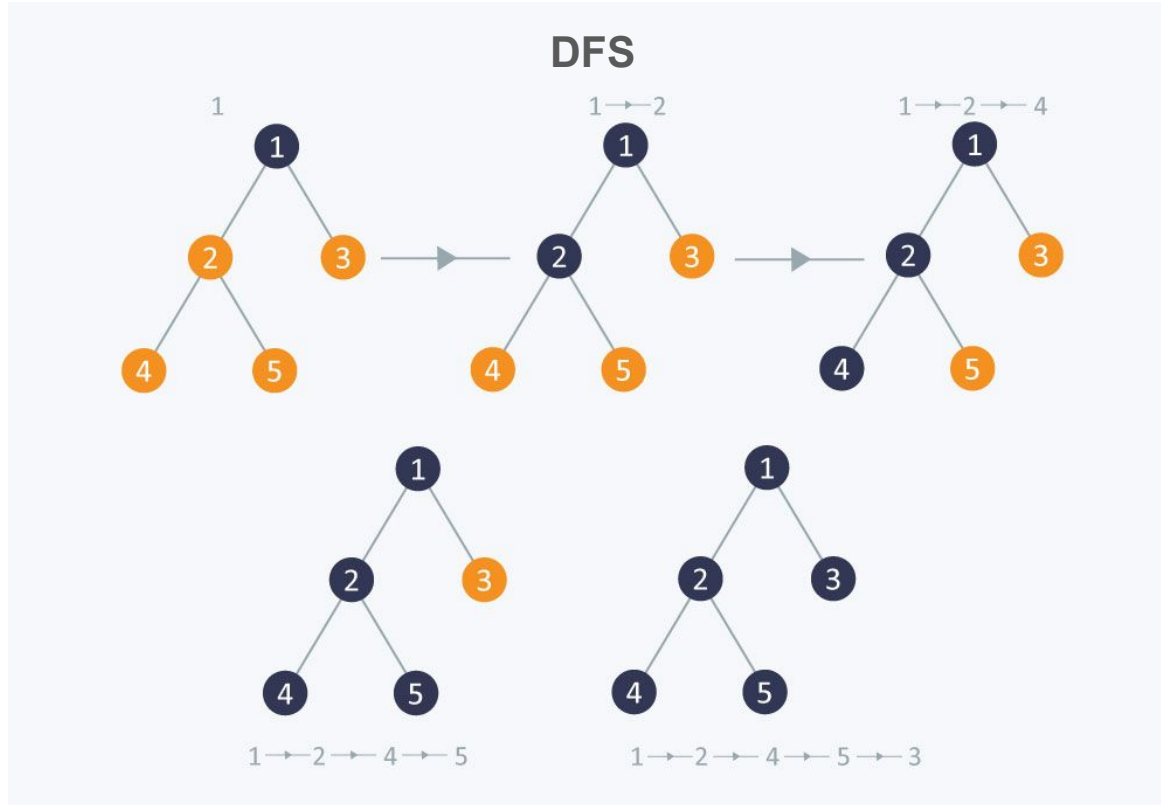
Center pixel has 2 neighbors → Edge



Node upsampling: Visvalingam-Whyatt algorithm



Graph traversal: Depth-first search (DFS)

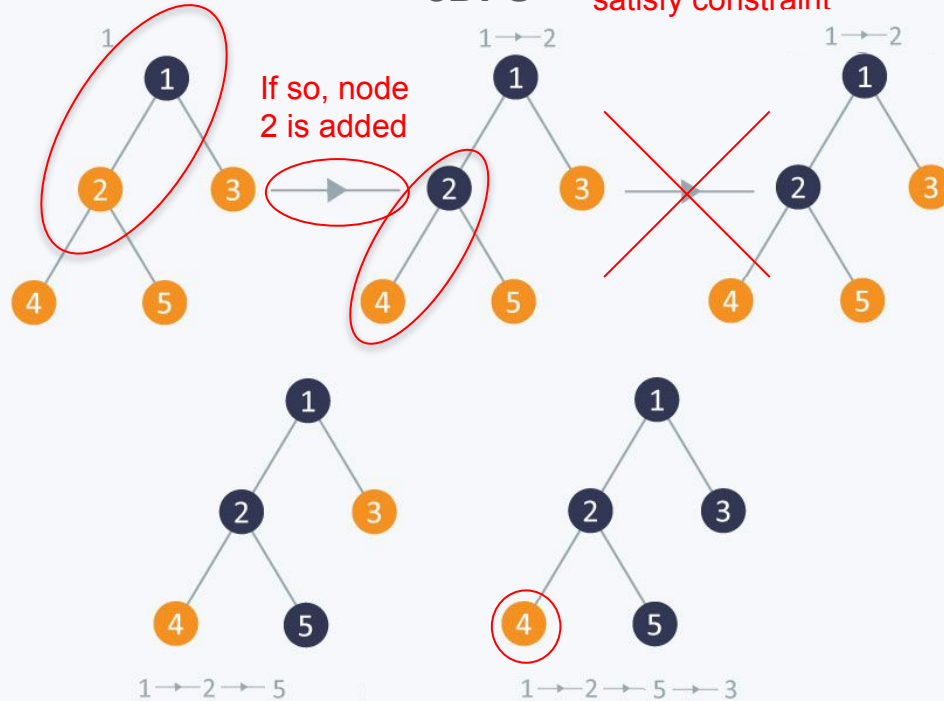


Graph traversal: Constrained Depth-first search (cDFS)

Does it satisfy the constraint?

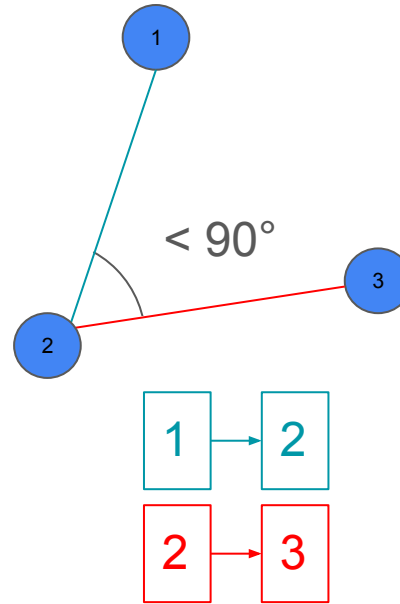
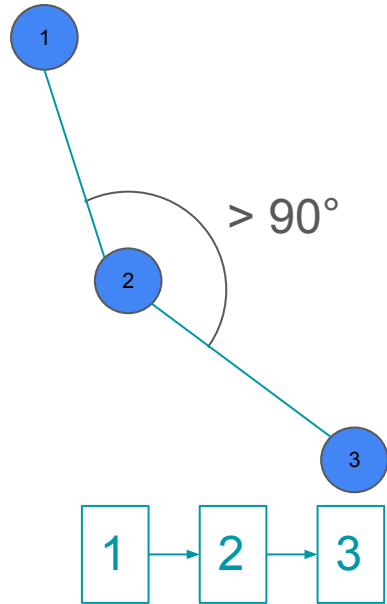
cDFS

Node 4 does not satisfy constraint



GraFT: cDFS – Introducing the angle constraint

- Angle between two edges must be equal or higher than 90°



GraFT Extension: Thickness and intensity constraints

- Thickness/intensity threshold
- If the thickness/intensity difference between 2 edge segments is above the threshold
→ Constraint violated

Threshold = 0.5

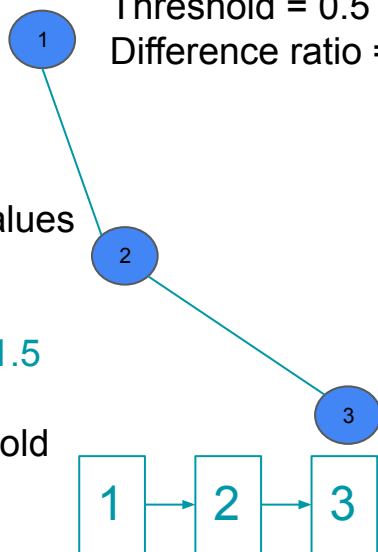
Difference ratio = $\text{abs}(\text{current value} - \text{last value}) / \text{last value}$

Thickness/Intensity values
of edges:

Last_value[1,2] = 2

Current_value[2,3] = 1.5

Ratio = 0.25 < Threshold

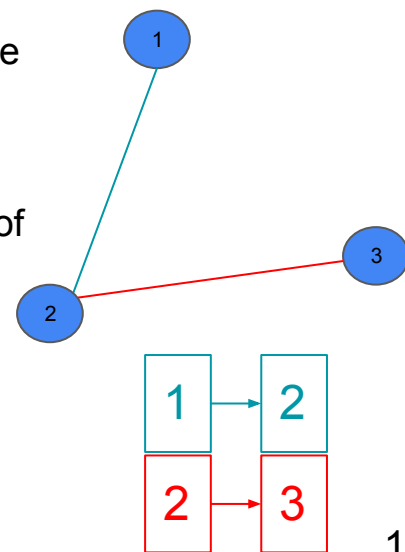


Thickness/Intensity values of
edges:

Last_value[1,2] = 2

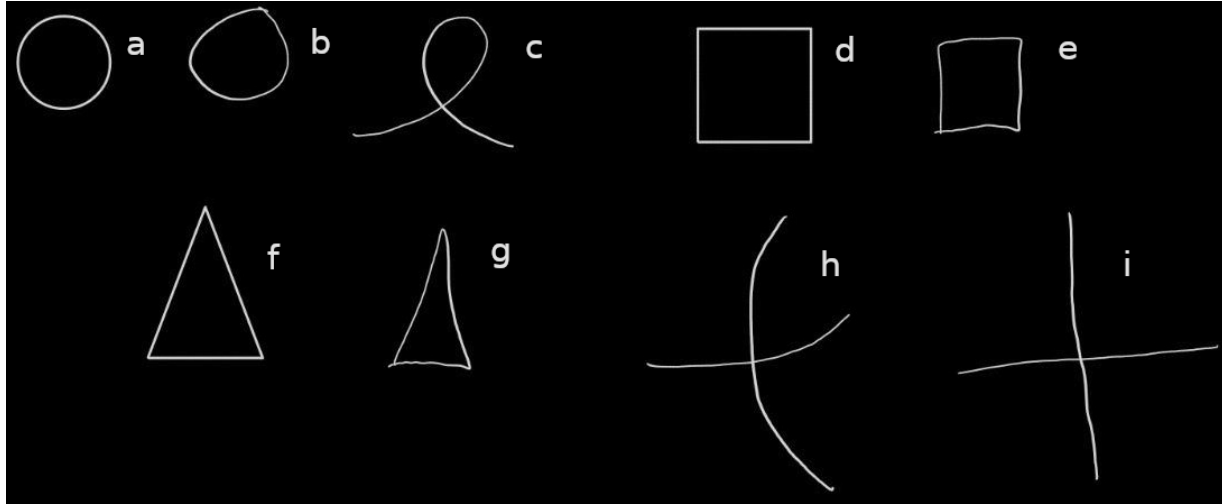
Current_value[2,3] = 0.5

Ratio = 0.75 > Threshold



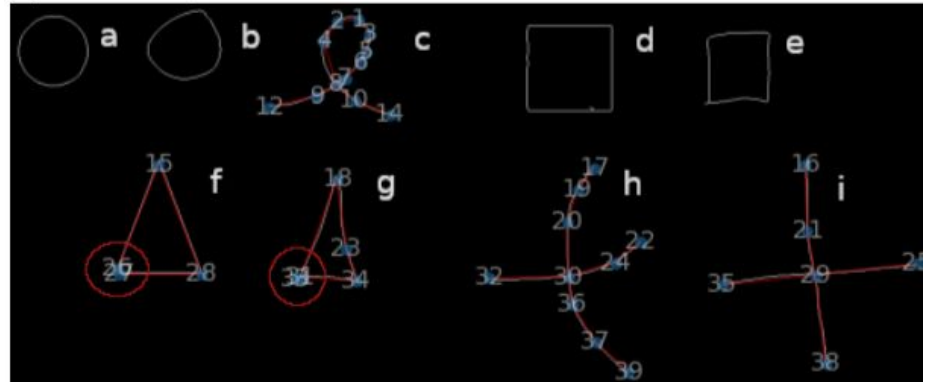
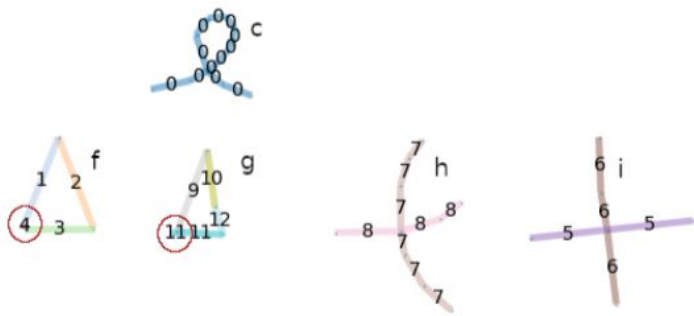
Performance test using a synthetic image

- 9 structures: 6 closed (a, b, d, e, f, g) shapes and 3 open shapes (c, h, i)



Original GraFT performance

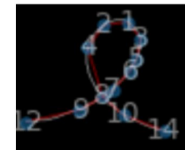
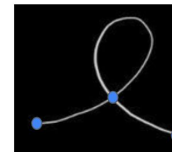
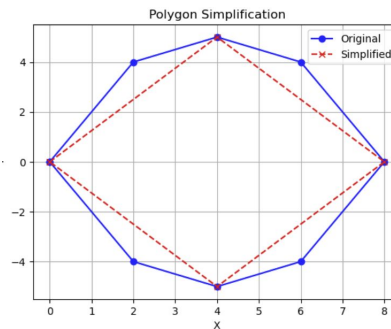
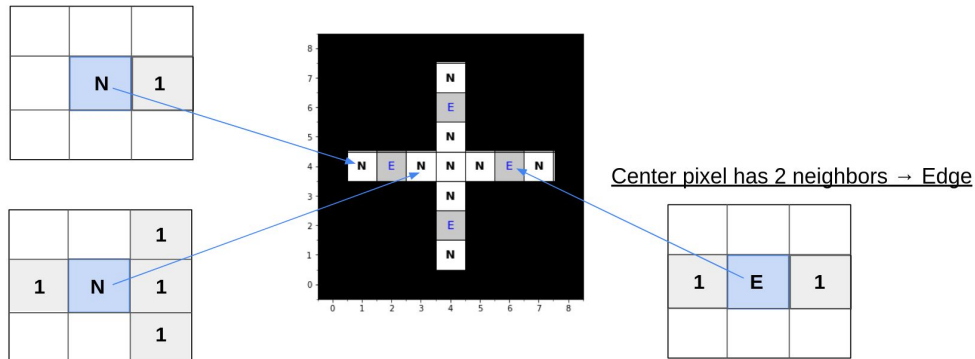
- Circles (a,b) and rectangles (d,e) were not traced → Closed shapes (nodes of degree 2)



Original GraFT: Closed shape limitation

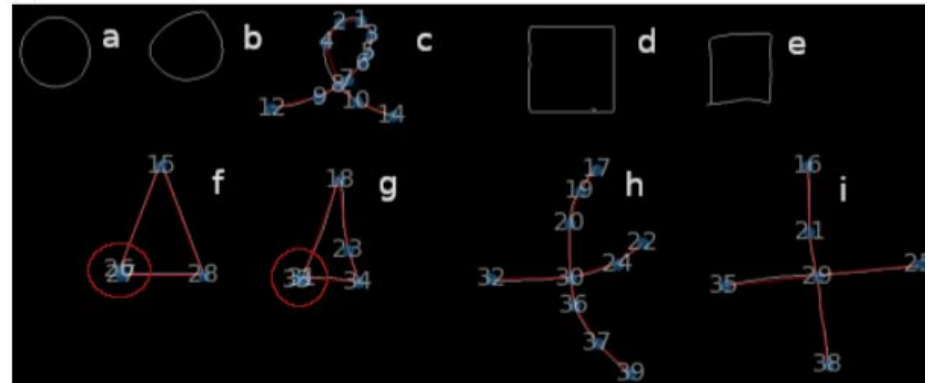
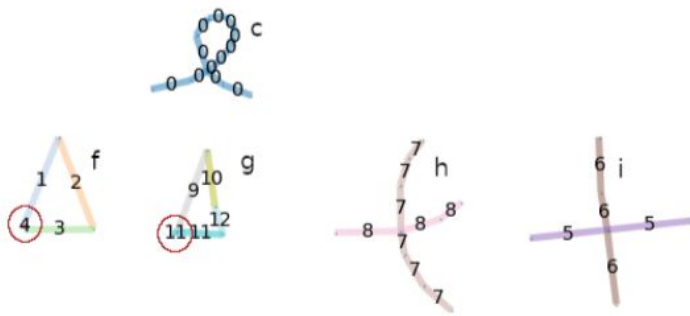
- Initial nodes are placed only on pixels with 1 or +3 neighbors
- Original VW algorithm only iterates through premarked nodes

Center pixel has 1 or 3 or more neighbors → Dangling Node

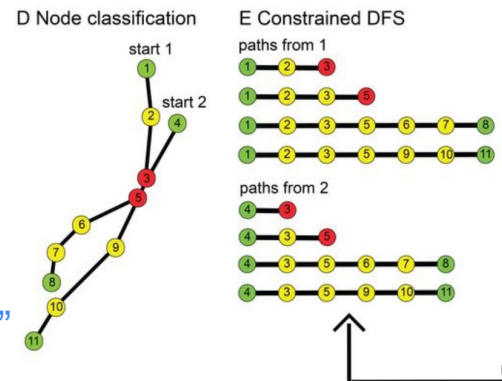


Original GraFT performance

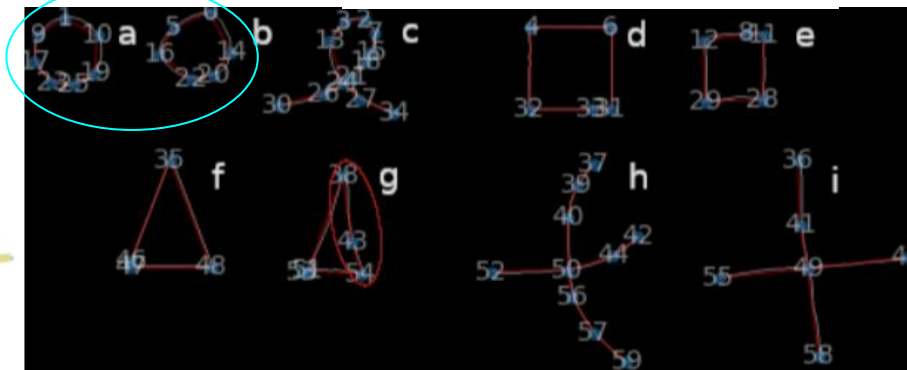
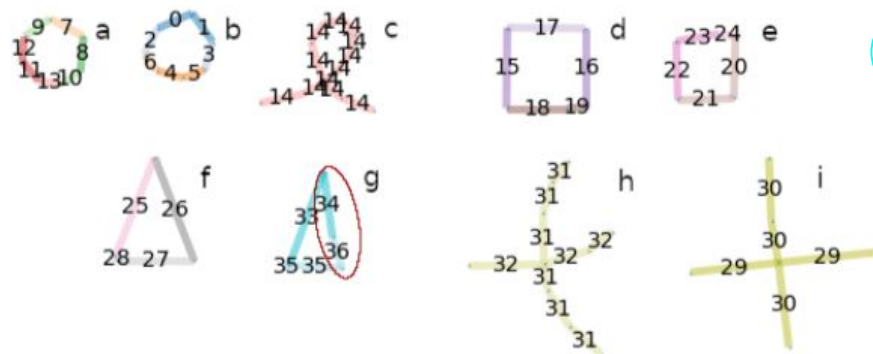
- Circles (a,b) and rectangles (d,e) were not traced → Closed shapes (nodes of degree 2)



Extended GraFT - Closed shapes

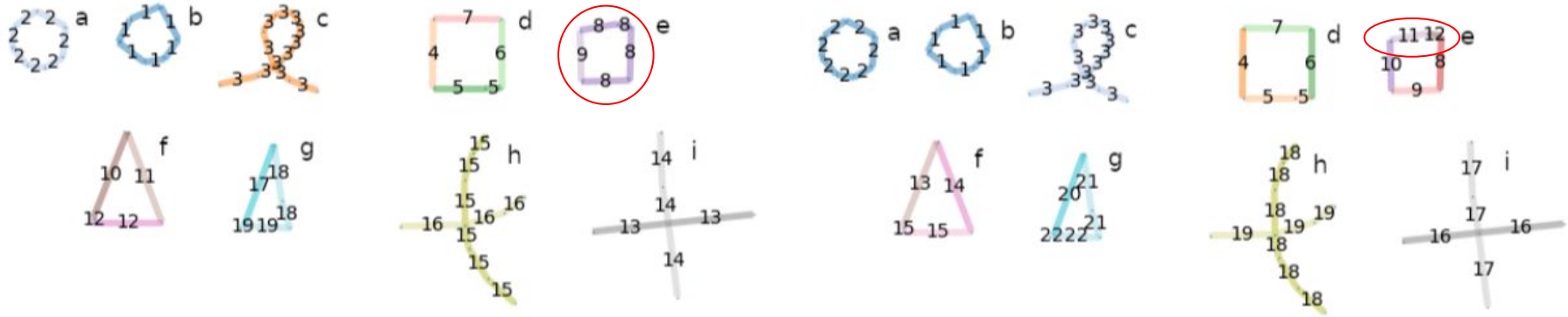


Nodes are not “dangling”



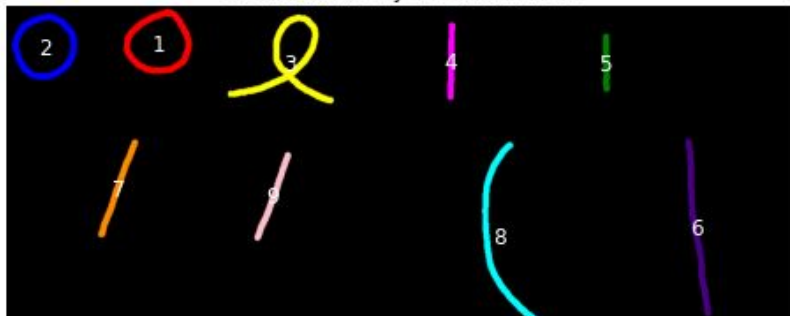
Extended GraFT: Dangling nodes – Thickness constraint

- Add thickness constraint → improvement in structure (e)

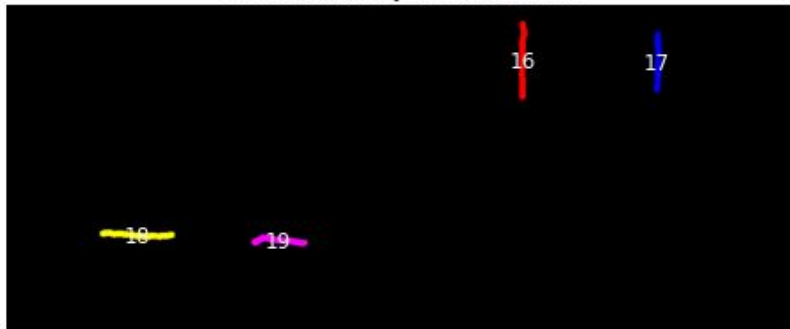


Performance quantification: Ground truth labels

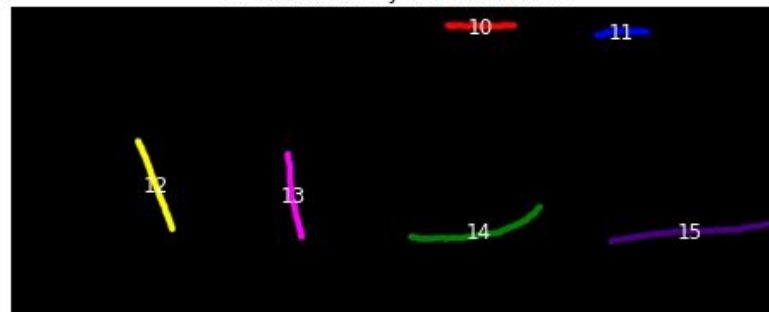
Ground Truth Layer 1 with Labels



Ground Truth Layer 3 with Labels



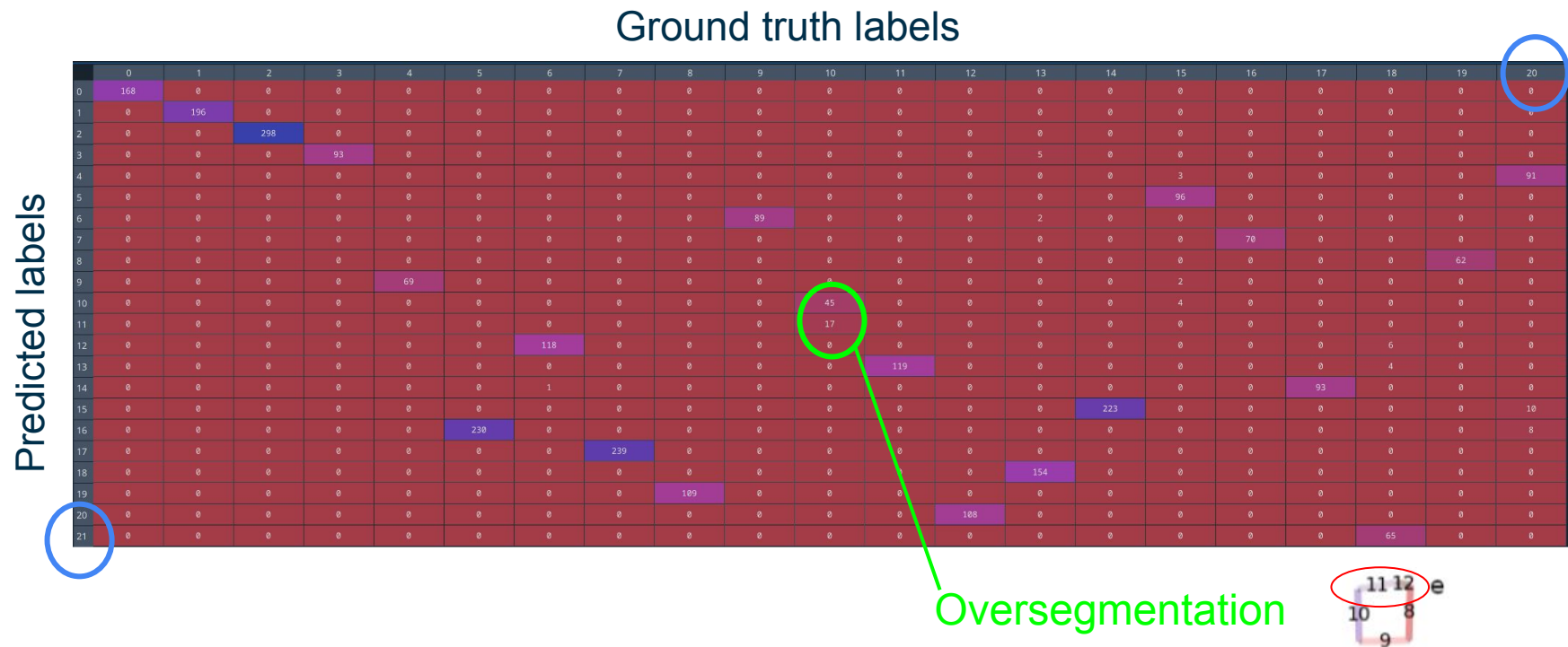
Ground Truth Layer 2 with Labels



Ground Truth Layer 4 with Labels



Performance quantification: Confusion matrix



Performance quantification: Metrics

Filament metrics:

true_positive: 21

false_positive: 1

false_negative: 0

true_negative: 440

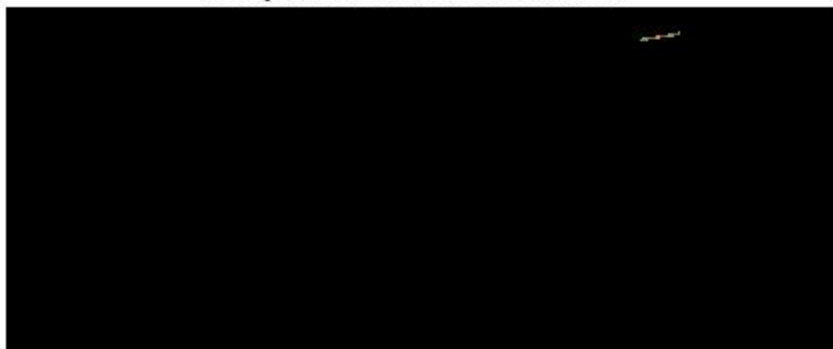
precision: 0.9545454545454546

recall: 1.0

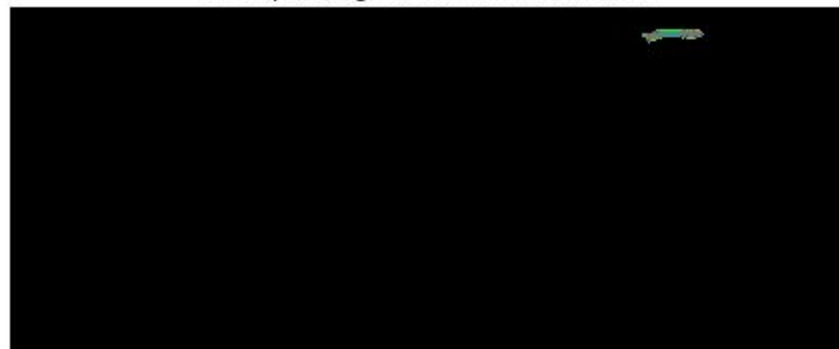
f1_score: 0.9767441860465117

mcc: 0.9759000729485331

Badly Matched Predicted Filaments

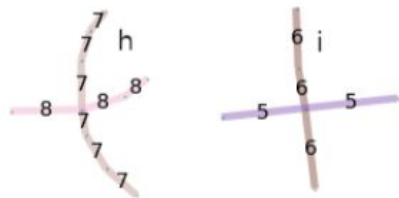


Corresponding Ground Truth Filaments



Key takeaways

- Original GraFT traces well simple overlaps



- Extension:
 - Considers closed shapes
 - Avoids over-segmentation in closed shapes
 - Thickness/intensity constraints increases accuracy by reducing under-segmentation
- Framework needs to be tested and tuned on real image data